

Roll-Over Proof Car

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Abstract—This paper presents the design and implementation of a roll-over proof car using an MPU6050 sensor and servo motors. The objective of this project is to enhance the safety of the vehicle by preventing potential rollover accidents. The MPU6050 sensor is utilized to detect the inclination angle of the car, while servo motors are employed to counteract the rolling motion. The experimental results demonstrate the effectiveness of the proposed system in maintaining the stability of the vehicle and mitigating the risks associated with roll-overs.

Keywords—Roll-over proof car, MPU6050 sensor, servo motors, stability, safety.

I. INTRODUCTION

Road accident is a major problem in our country. Any accident might result in people dying, damage to vehicles, property, and injured accident victims requiring medical treatment. Now, suppose a massive car accident has occurred in the middle of a busy road. The car is damaged heavily and the people in it have also received some severe injuries. So, someone calls an ambulance for the immediate treatment of injured humans and a tow truck for clearing the accident spot. But both the help doesn't reach the spot in time resulting in unfortunate death of the injured people and a large-scale traffic jam. Therefore, the solution that we proposed is to design and implement a roll-over proof car system. The system aims to detect the possibility of a roll-over event and activate the servo motor to stabilize the car by adjusting its centre of gravity.

II. LITERATURE SURVEY

Bulletproof vehicles (BPVs) are specialized automobiles made by adding interior steel compartments to already-existing automobiles. The safety of the passengers and maintaining vehicle manoeuvrability are the main priorities in BPV design [1]. However, there are difficulties in producing BPVs, including finding appropriate vehicle rollover models, choosing pertinent parameters, contrasting current models, and developing new models especially suited to BPVs in developing nations. Additionally, the process of

retrofitting existing vehicles with steel compartments can be time-consuming and costly. Furthermore, ensuring that the added weight of the steel compartments does not compromise the vehicle's performance and fuel efficiency poses another challenge in BPV design. To analyse the rollover tendencies in BPVs, this research aims to investigate and focus on the parameter selection process. The study also suggests a method for avoiding potential rollovers brought on by risky driving practices. When creating rollover models for these specialized bulletproof vehicles, various scenarios are taken into account. The additional armoured protection adds weight to the vehicle, which is an important factor, and poses unique difficulties for BPV manoeuvrability. More research is needed in this area because the investigation of rollover tendencies in BPVs goes beyond what can be captured by conventional vehicle models. The study shows that an unbalanced weight distribution in BPVs can cause hidden instability tendencies, calling for additional research. The ultimate objective is to improve BPV stability and safety to meet the needs of these specialized vehicles. One important factor to consider in the research is the effect of varying load conditions on BPV stability. Understanding how different weight distributions impact manoeuvrability will be crucial in developing effective solutions. Additionally, the unique challenges posed by off-road terrain and uneven surfaces should also be considered to ensure that BPVs can safely navigate these environments.

A roll-over prevention system for commercial vehicles using an additional sensorless function for the electronic brake system (EBS) is designed [2]. It identifies roll-over accidents as a significant safety concern and proposes a solution that utilizes existing EBS sensors and actuators to detect and prevent roll-over incidents. A patented Roll-Over Prevention (ROP) system is introduced that triggers the brake or throttle system if lateral acceleration surpasses a certain threshold. Additionally, the Drive Stability Control (DSC) system is presented as an extra function of the EBS, using additional sensors to optimize vehicle stability during challenging manoeuvres. The research highlights the importance of these enhancements for vehicle safety and

emphasizes the effectiveness of the proposed systems in preventing roll-over accidents.

A fault-tolerant rollover prevention system aims to develop a fault-tolerant rollover prevention system to improve the performance properties of heavy vehicles and reduce the risk of rollovers vehicles based on the Linear Parameter-Varying (LPV) method by using a fault-tolerant reconfigurable controller, which includes active anti-roll bars, active suspensions and an active brake, has been proposed [3]. The LPV approach allows for robust control design by considering the varying parameters and uncertainties present in the system. By incorporating fault-tolerant mechanisms, the proposed system enhances vehicle safety by mitigating the risks of rollovers even in the presence of faults. This project mainly focusses on designing a combined controller that uses active suspensions and active anti-roll bars all the time for improving passenger comfort, road holding and guarantee the suspension working space, activating the controlled braking system only when the vehicle comes close to rolling over. On dangerous situation the active brake system must generate unilateral brake forces in order to reduce the risk of rollovers. By incorporating fault-tolerant mechanisms, the proposed system enhances vehicle safety by mitigating the risks of rollovers even in the presence of faults.

Autoliv has developed a new rollover protection system called Advanced Roll Over System by Autoliv (AROSA). AROSA is designed to trigger protective elements such as head bags, side bags, and belt pretensioners at the optimum moment in a rollover accident [4]. AROSA was developed after Autoliv conducted extensive research into the causes of rollover accidents. The system uses a combination of sensors to measure the vehicle's rotational speed, lateral acceleration, and vertical acceleration. AROSA also uses the vehicle's speed as an additional input to ensure that the protective elements are triggered in time. AROSA is doubly safe, meaning that it has two separate systems that must both agree before the protective elements are triggered. AROSA can be integrated into the existing safety system of a vehicle.

III. SYSTEM ARCHITECTURE

The roll-over proof car system consists of the following main components: an MPU6050 sensor, servo motors, a microcontroller unit (MCU), and a power supply. The system utilizes the MPU6050 sensor, which combines both accelerometer and gyroscope functionalities, to detect the inclination angle of the car. An Arduino microcontroller (the MCU) is used to interface with the MPU6050 sensor and servo motor. It processes the sensor data and controls the servo motor accordingly. When the MPU6050 detects an excessive inclination angle, the servo motor counterbalances it by adjusting two upright wooden logs connected with two servos that are positioned at the two sides of the car.

A. MPU 6050: A Robust Inertial Sensing Approach

The MPU6050 is a widely used inertial measurement unit (IMU) that combines a three-axis accelerometer and a three-axis gyroscope in a single chip. Its small form factor, low power consumption, and excellent performance make it an ideal choice for measuring angular velocity, acceleration, and orientation changes in real-time.

The MPU6050 mechanism in the roll-over proof car is designed to continuously monitor the vehicle's tilt angles and detect potential roll-over situations. By accurately measuring pitch and roll angles, the system can make proactive adjustments to maintain vehicle stability during critical manoeuvres. The MPU6050 communicates with the microcontroller, which processes the sensor data and determines the appropriate corrective action.

B. Servo Motor System

The servo motor system plays a crucial role in actively countering the tipping motion of the vehicle. In this project, two servo motors are employed, each connected to a wooden log positioned opposite the other. When the MPU6050 sensor detects an inclination exceeding a predefined threshold, the servo motors are engaged to roll back the logs. This action applies an opposing force to the tipping direction, effectively stabilizing the vehicle. The servo motors used in this implementation provide precise control over the rotation angle, allowing for quick and accurate response to detected roll-over risks. By leveraging the servo motor system, the roll-over proof car enhances stability and minimizes the possibility of accidents.

C. Microcontroller Unit (Arduino)

To coordinate the sensor data acquisition and servo motor control, an Arduino microcontroller unit is utilized. The Arduino acts as the central processing unit, receiving inputs from the MPU6050 sensor and generating appropriate commands for the servo motors. It employs a robust control algorithm that analyzes the inclination data and triggers the servo motors when necessary. The Arduino's programmability and versatility enable the implementation of complex decision-making algorithms, enhancing the roll-over prevention system's effectiveness. Moreover, it provides convenient interfacing capabilities with the MPU6050 sensor and servo motors, ensuring seamless integration of the various components.

IV. SENSOR INTEGRATION AND DATA PROCESSING

When the sensor is kept in idle condition in the ground, there is no reading in the x or y direction (i.e., in the two horizontal directions of the plane where the car is standing) but we can see a force close to 10m/s^2 in z axis (vertical axis to the plane of the car), which is because there is a force $1g$ due to gravity. Now we will use that force and break it into vectors and use them to find the angles [6].

Now here the sensor is making an angle θ with the horizontal plane. Now we are drawing a vector triangle to help us find the components. Drawing the vector components we can see that the triangle is also making angle θ . Now we know that $\tan$ of θ is perpendicular upon base. Here we can see that perpendicular is acceleration in x and the base is acceleration in z. Therefore $\tan \theta$ is acceleration in x divided by acceleration in z. Again if we take tilt angle in y direction and take that angle as ϕ then for that the formula will be acceleration in y divided by acceleration in z.

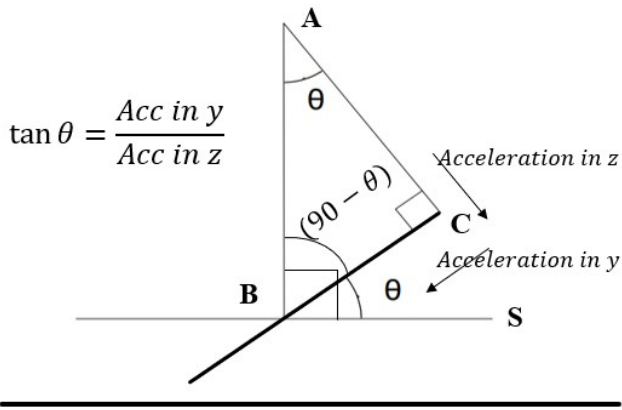


Fig. 1. Tilt Angle

V. SERVO MOTOR CONTROL AND CONTROLLING THE SYSTEM

If we want to rotate an object at some specific angles or distance, then we use a servo motor. It is just made up of a simple motor which runs through a servo mechanism.

Now as we have extracted the angle from the MPU 6050 we can use it to know if the car has rolled over or not. We have taken the threshold value for accident to be 60 degrees offshoot.

Now we will use servos to give leverage for turning it up. We have attached two wooden logs with the servos. Now we have made two servo objects for two directions, X servo and Y servo. If the angle offshoots by 60 degrees in the right side, that is the value of the angle increases beyond 60 degrees then the right servo is moved. In the same way, if it offshoots in the left side that is the angle calculated decreases below -60 degrees then the left servo is moved.

To turn the vehicle upright, the wooden piece is turned 90 degrees in the direction of the roll then it is again taken back to its previous position.

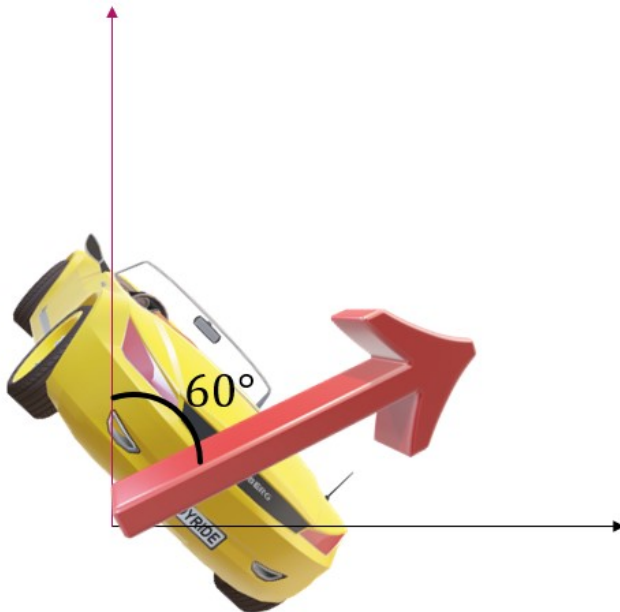


Fig. 2. Threshold Angle for Accident

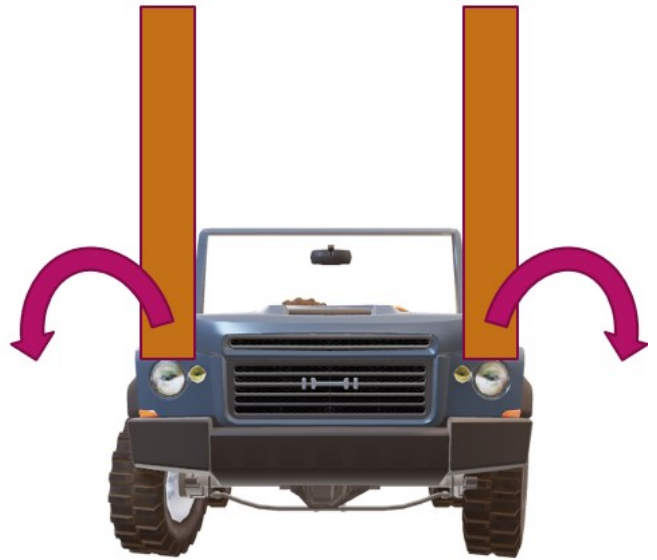


Fig. 3. Position of the logs

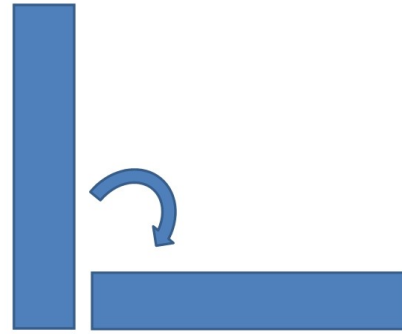
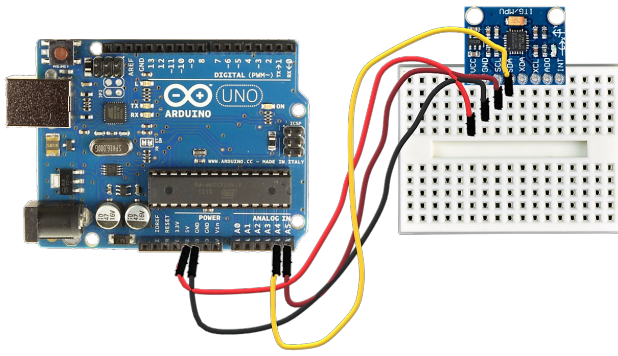


Fig. 4. Path of the logs

VI. ARDUINO PROCESSING

A. Connections

The MPU 6050 communicates with the Arduino through the Inter-Integrated Circuit (I²C) protocol [5]. Voltage Common Collector (VCC) is connected to the 5V output of the Arduino. Next, the ground (GND) of the Arduino is connected to the GND of the MPU 6050. Now, the I²C lines are set up. We will connect the Serial Clock (SCL) pin to Arduino's analog pin 5 (SCL). Serial Data (SDA) is a I²C Data pin used for both transmit and receive. We will connect the SDA pin to the Arduino's analog pin 4 (SDA). The servo is also attached with a digital pin in the Arduino.



- Red is connected from 5V pin to GND of MPU6050
- Black is connected from Ground pin to GND of MPU6050
- Yellow is connected from A5 to SCL of MPU6050
- White is connected from A4 to SDA of MPU6050

Fig. 5. Connections

B. Arduino Coding

In the code, we first include the necessary libraries for Wire communication, Servo motor control, and MPU6050 sensor. We create objects for the MPU6050 sensor and the servo motors. We also define the necessary pins and variables for calibration, servo motor control, tilt threshold, and adjustment factor.

In the `setup()` function, we have initialized the serial communication, the MPU6050 sensor, calibrated it, and attach the servo motors to the specified pins.

In the `loop()` function, we have read the accelerometer data from the MPU6050 sensor and calculate the tilt angles in degrees. If the absolute value of the tilt angle exceeds the defined threshold, we calculate the adjustment angle and the target positions for the servo motors. Finally, we move the servo motors to the target positions. The tilt angles are also printed to the serial monitor for debugging purposes.

VII. CONCLUSION AND FUTURE ASPECTS

This paper presented the design and implementation of a roll-over proof car system utilizing an MPU6050 accelerometer and servo motors. The proposed system effectively detected the inclination angles of the vehicle and applied corrective forces through servo motors to prevent

excessive rolling and improve stability. The experimental results validated the feasibility and efficiency of the system in mitigating the risks associated with roll-over accidents. Future work may involve refining the system for real-world applications and integrating additional safety features to enhance overall vehicle safety. This idea is not only for cars, we can implement this idea in other vehicles also.

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REFERENCES

- [1] S. G. H. Naqvi et al., "The rollover prediction and prevention of bullet proof vehicles for improved stability," *Life Sci. J.*, vol. 10, no. 4s, pp. 209-214, 2013. ISSN: 1097-8135.
- [2] L. Palkovics et al., "Roll-over prevention system for commercial vehicles – Additional sensorless function of the electronic brake system," *Veh. Syst. Dyn.*, vol. 32, pp. 4-5, 285-297, 1999, doi:10.1076/vesd.32.4.285.2074.
- [3] Gáspár, P. and Bokor, J. (2006) 'A fault-tolerant rollover prevention system based on an LPV method', *Int. J. Vehicle Design*, Vol. 42, Nos. 3/4, pp.392 – 412.
- [4] J. Mayer et al., "New rollover protection system," *ATZ Worldwide*, vol. 103, no. 7-8, pp. 6-7, 2001. doi:10.1007/BF03226797.
- [5] Last Minute Engineers, "In-depth: Interface MPU6050 Accelerometer & Gyroscope sensor with Arduino," Last Minute Engineers, <https://lastminuteengineers.com/mpu6050-accel-gyro-arduino-tutorial/>.
- [6] P. McWhorter, "9-Axis Inertial Measurement Unit (IMU) | Technology Tutorials," *Toptechboy*. <https://toptechboy.com/arduino-based-9-axis-inertial-measurement-unit-imu-based-on-bno055-sensor/>.